Mastering Modern LLMs

The definitive engineering guide to Fine-Tuning, Quantisation, and Alignment.

From raw base models to efficient, safe, and production-ready generative AI.

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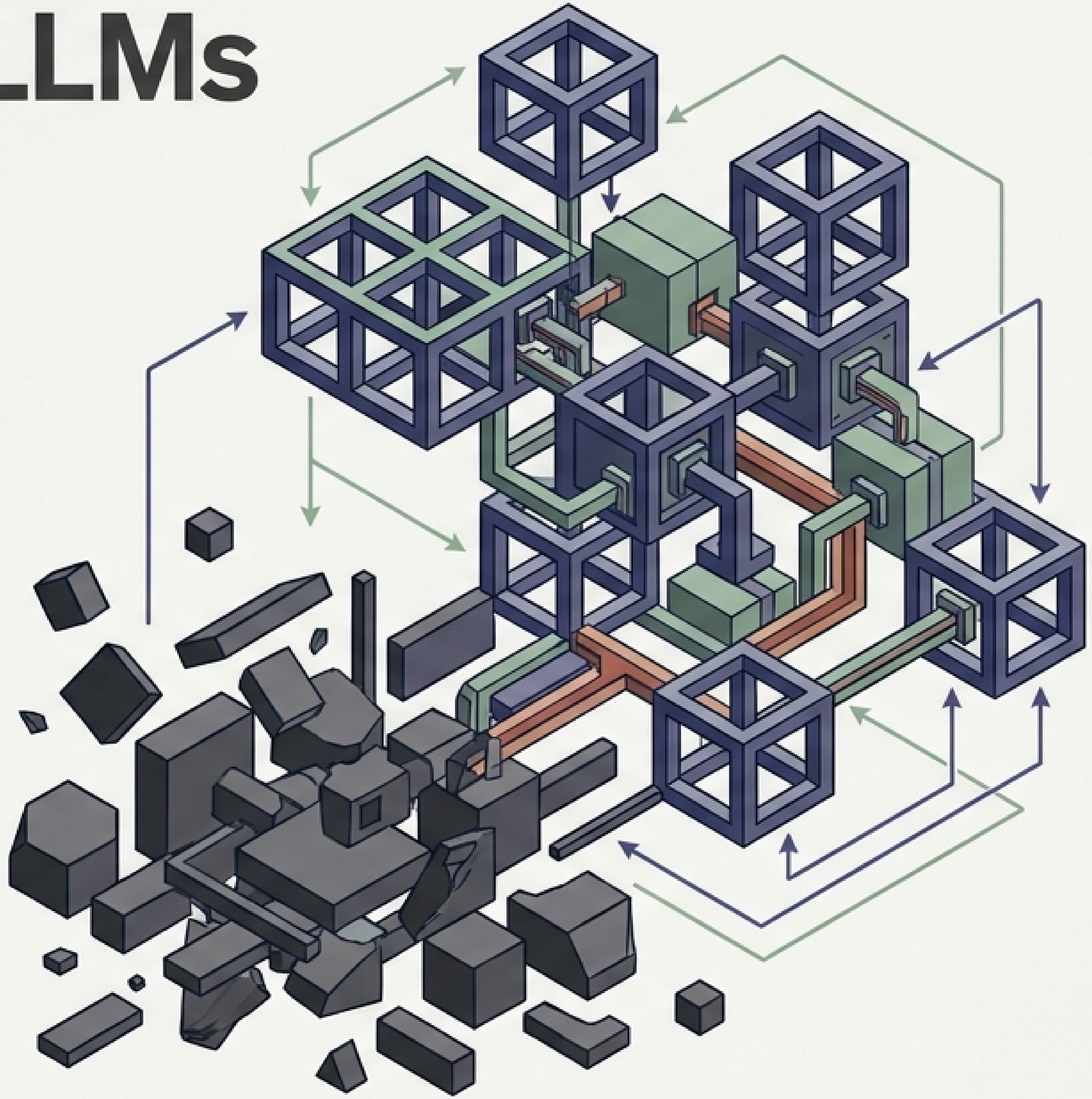
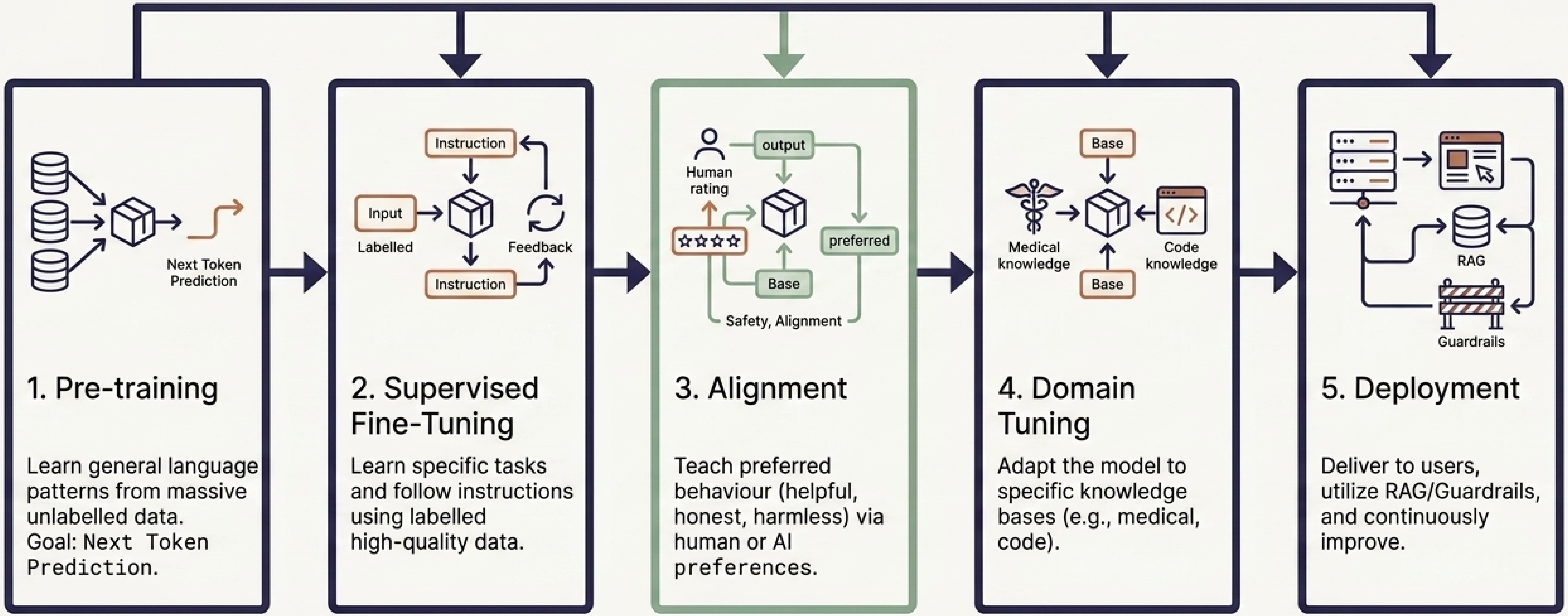


The Lifecycle of a Large Language Model



The High Cost of Full Fine-Tuning

The Problem

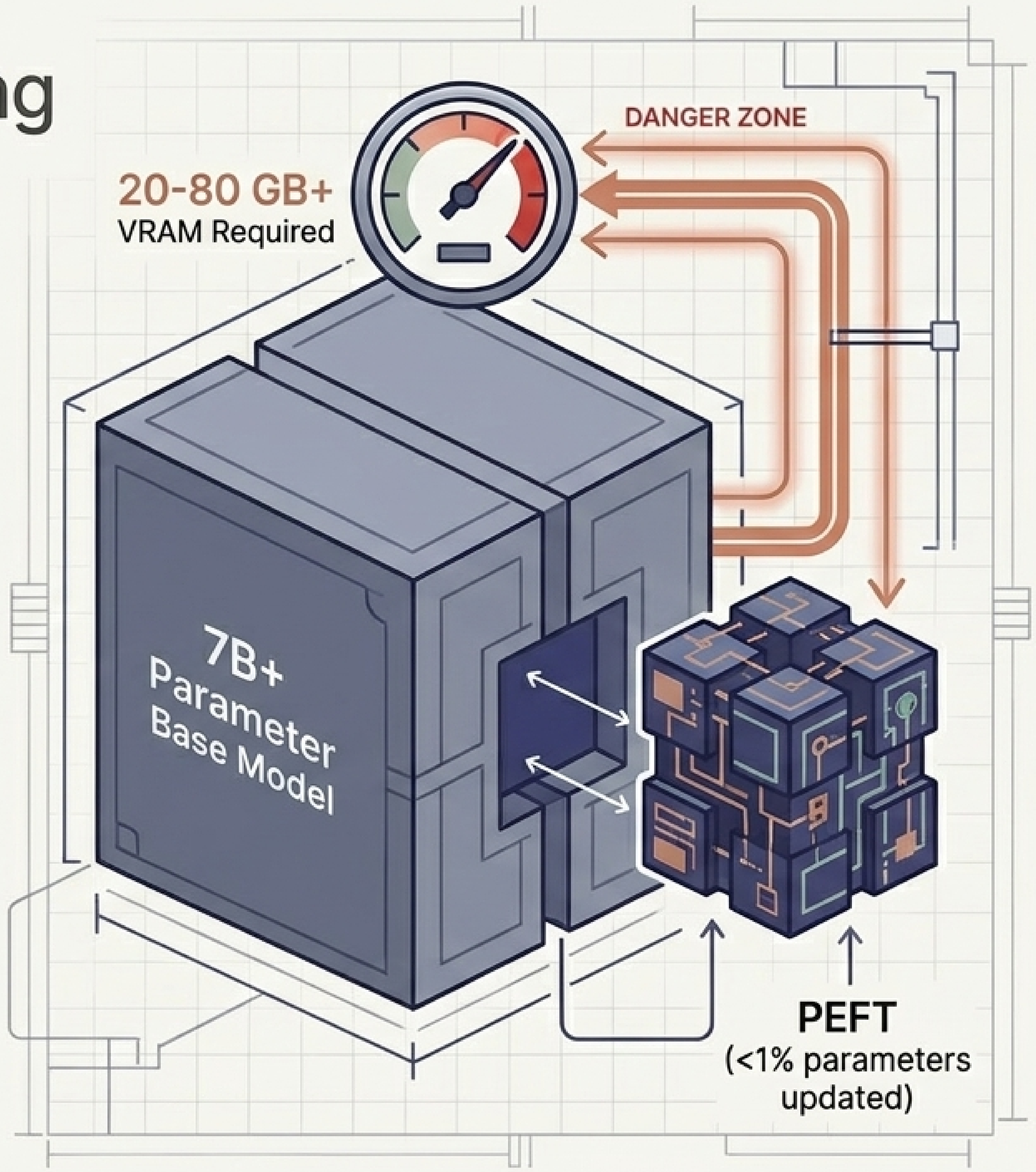
Full fine-tuning requires updating billions of parameters, causing massive GPU memory demands, slow training times, and risking Catastrophic Forgetting.

The Solution: PEFT

Parameter-Efficient Fine-Tuning (PEFT) freezes original weights and trains only a tiny fraction of new parameters.

Benefits

Faster training, cheaper hardware (consumer GPUs), and smaller, swappable model checkpoints.

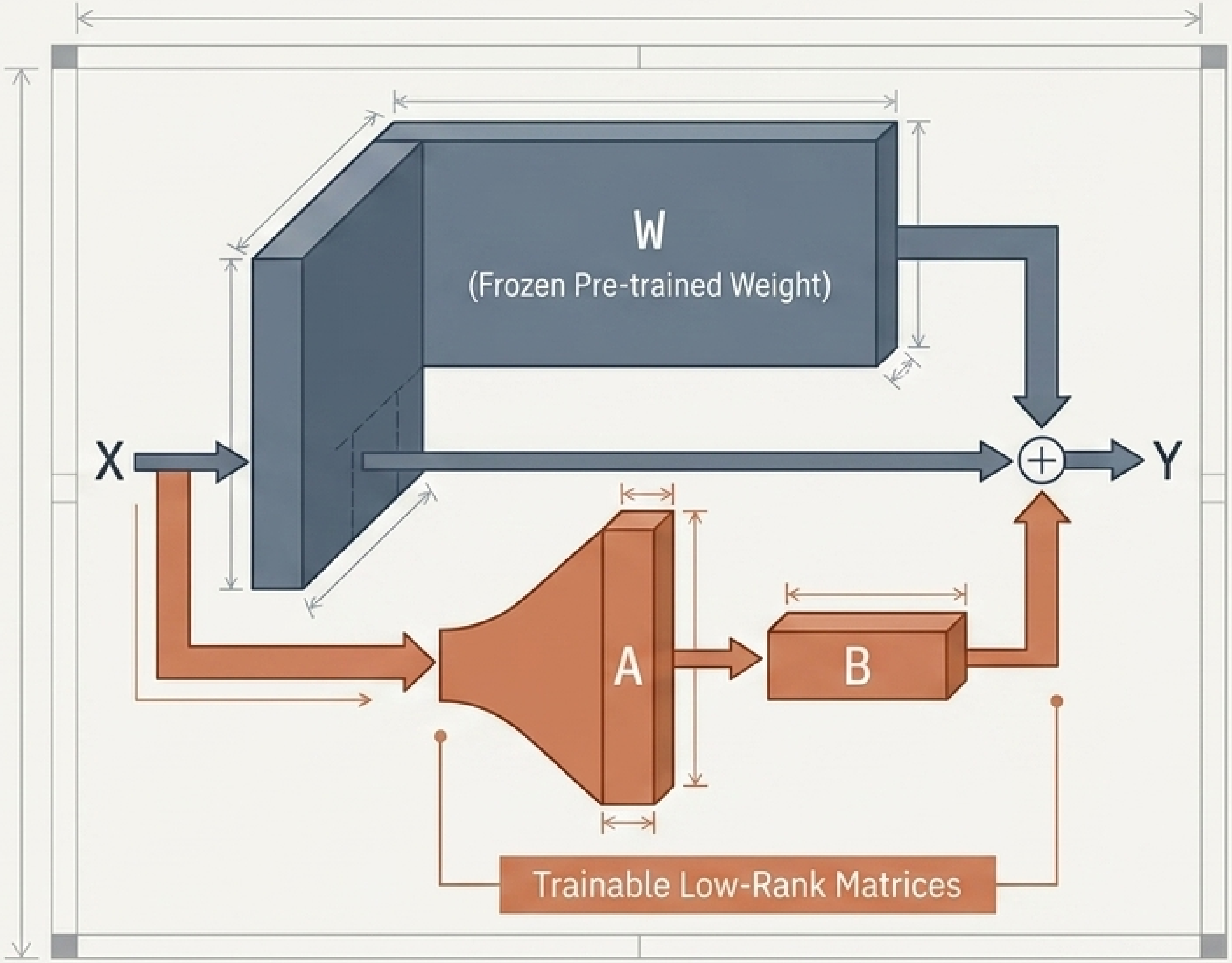


Bypassing the Bottleneck with LoRA

Core Concept

Low-Rank Adaptation (LoRA) assumes task-specific updates lie in a low-dimensional space. Instead of learning the full ΔW, it learns two small low-rank matrices.

The Equation: $Y = X(W + BA)$



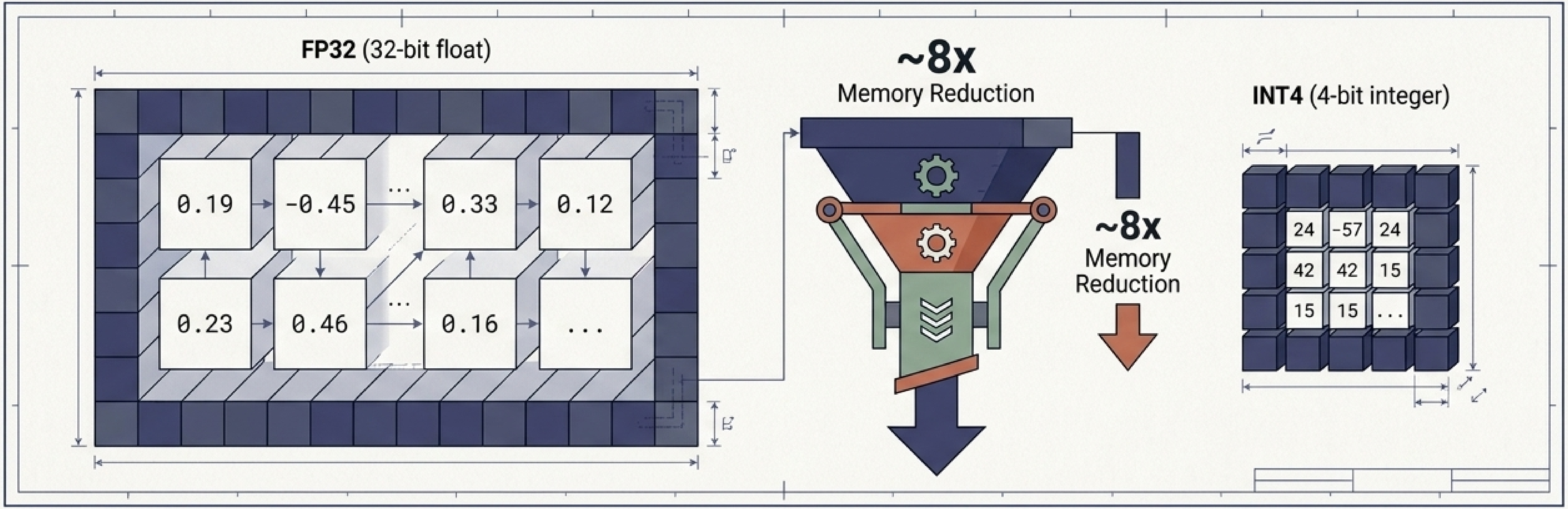
Key Hyperparameters

Rank (r): Controls the size of the adapter bottleneck. Common: 4, 8, 16, 32.

Alpha (α): The scaling factor applied to BA.

Dropout: Regularization applied during training (0.05 - 0.1).

Squeezing Model Weights via Quantisation



Definition

Reducing the numerical precision of weights to dramatically decrease model size and speed up inference.

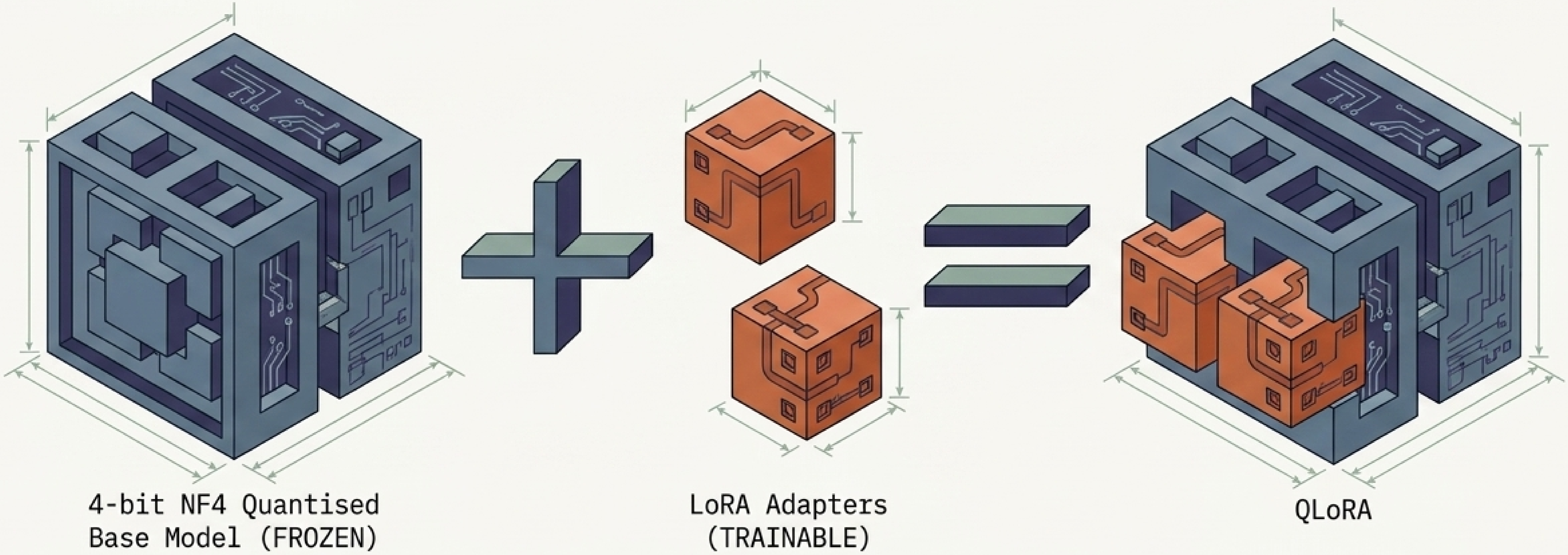
The Impact

Moving from FP32 to INT4 yields roughly an 8x memory reduction, allowing massive models to run on edge devices.

Two Primary Approaches

- **PTQ (Post-Training)**: Quantise the model after training is complete.
- **QAT (Quantisation-Aware)**: Train the model while simulating quantisation for higher accuracy.

QLoRA: The Ultimate Efficiency Combo



The Breakthrough

By combining a quantised base model with trainable LoRA adapters, QLoRA enables fine-tuning massive 65B parameter models on a single consumer GPU without catastrophic quality loss.

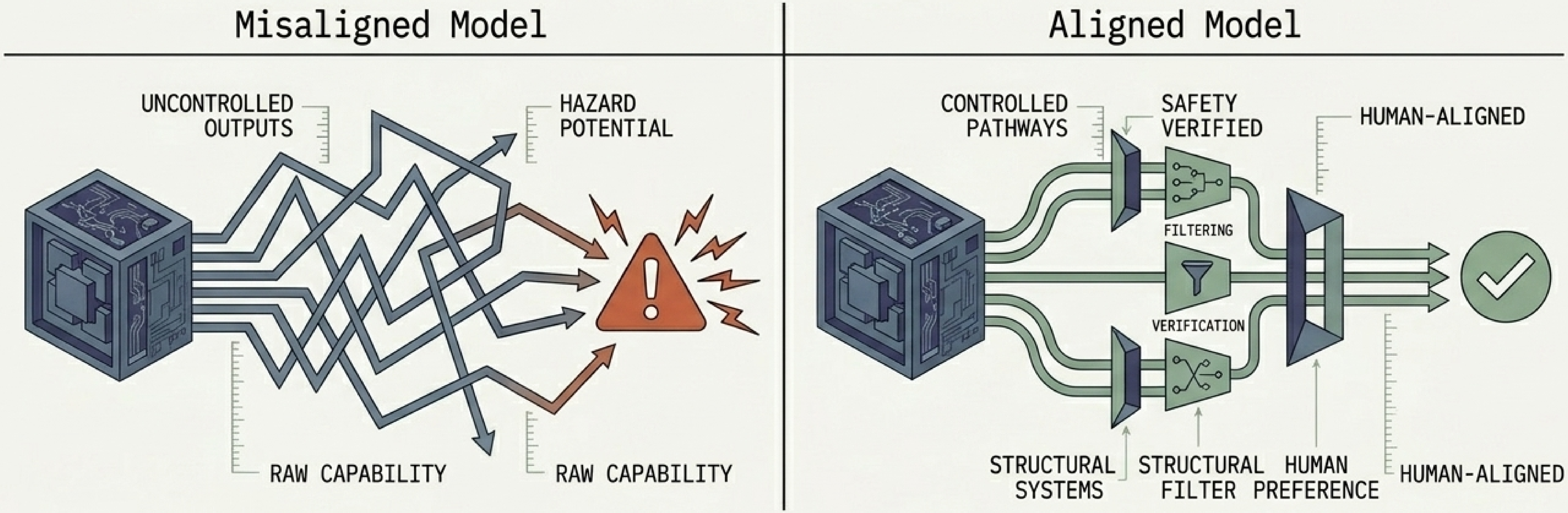
Specialised Techniques

- **NF4 (NormalFloat4)**: A special 4-bit data type optimised theoretically for the distribution of LLM weights.
- **Double Quantisation**: Quantising the quantisation constants themselves to squeeze out even more memory savings.

The Fine-Tuning Efficiency Matrix

	Full Fine-Tuning	PEFT (General)	LoRA	QLoRA
Changes to Base	All Weights	Subset of Weights	Frozen + Adapters	Quantised & Frozen + Adapters
Trainable Params	100% (7B+)	< 1%	< 1% (~1-10M)	< 1%
GPU Memory	Massive (26GB+)	Low	Very Low	Lowest (~10GB)
Speed	Very Slow	Fast	Fast	Fast
Ideal For	Foundational pivots	Efficient general tuning	Task adaptation	Huge models on limited hardware

Bridging the Gap Between Capability and Intent



The Alignment Imperative

A raw, pre-trained model is a statistical parrot–it predicts tokens, indifferent to truth, safety, or human preference. Alignment forces the model to match human values.

The HHH Goal

Aligned AI must be Helpful, Honest, and Harmless.

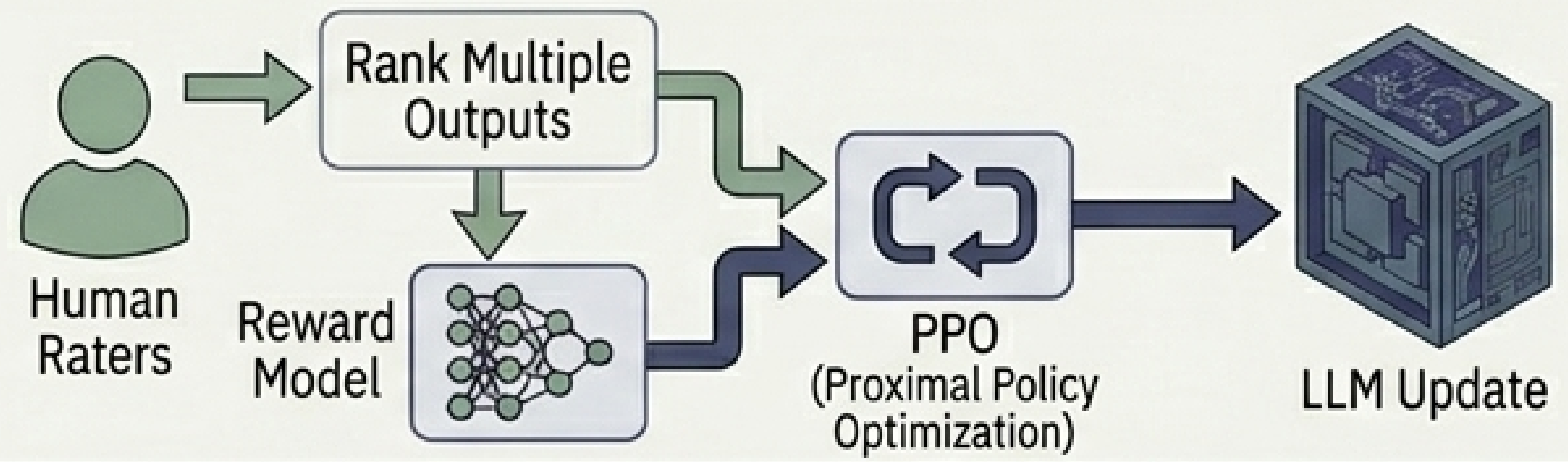
- **Helpful**: Addresses user intent constructively.
- **Honest**: Provides accurate, truthful information.
- **Harmless**: Avoids generating dangerous or toxic content.

The Tradeoff

The 'Alignment Tax'. Pushing safety guardrails too high can restrict a model's creativity, usefulness, and logic capabilities.

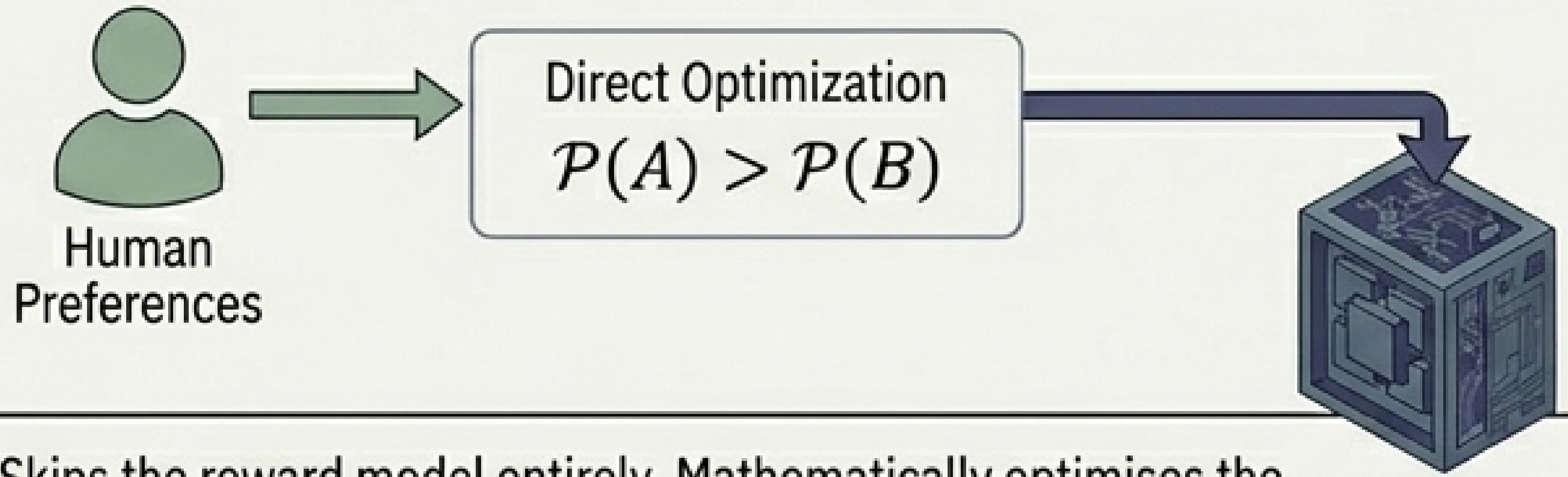
Mapping the Modern Alignment Ecosystem

1. RLHF (Reinforcement Learning from Human Feedback)

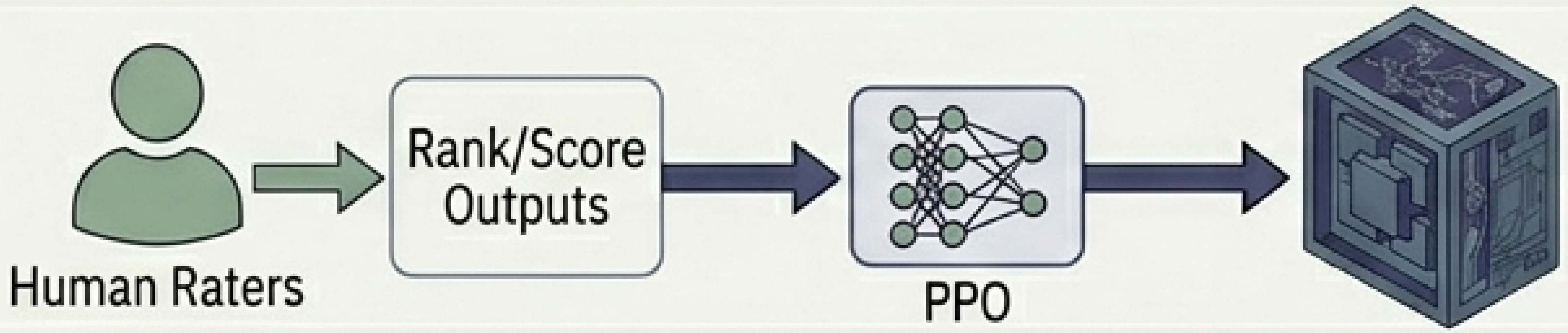


Humans rank multiple outputs. A Reward Model learns these preferences, which PPO uses to update the LLM.

Track 2. DPO (Direct Preference Optimisation)

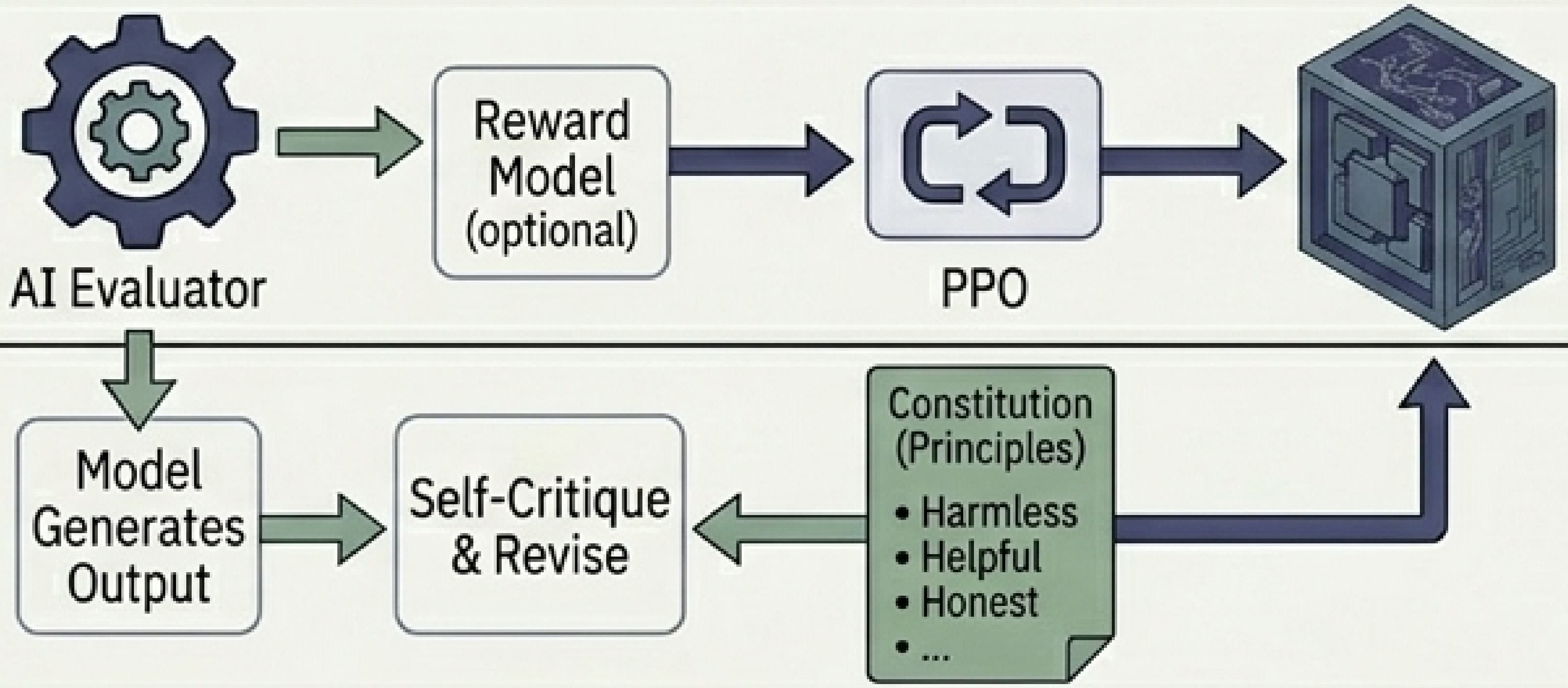


Skips the reward model entirely. Mathematically optimises the LLM directly on the human preference data. Simpler and stable.



Track 3: RLAIF Replaces:
Replaces expensive human raters with an AI evaluator. Highly scalable.

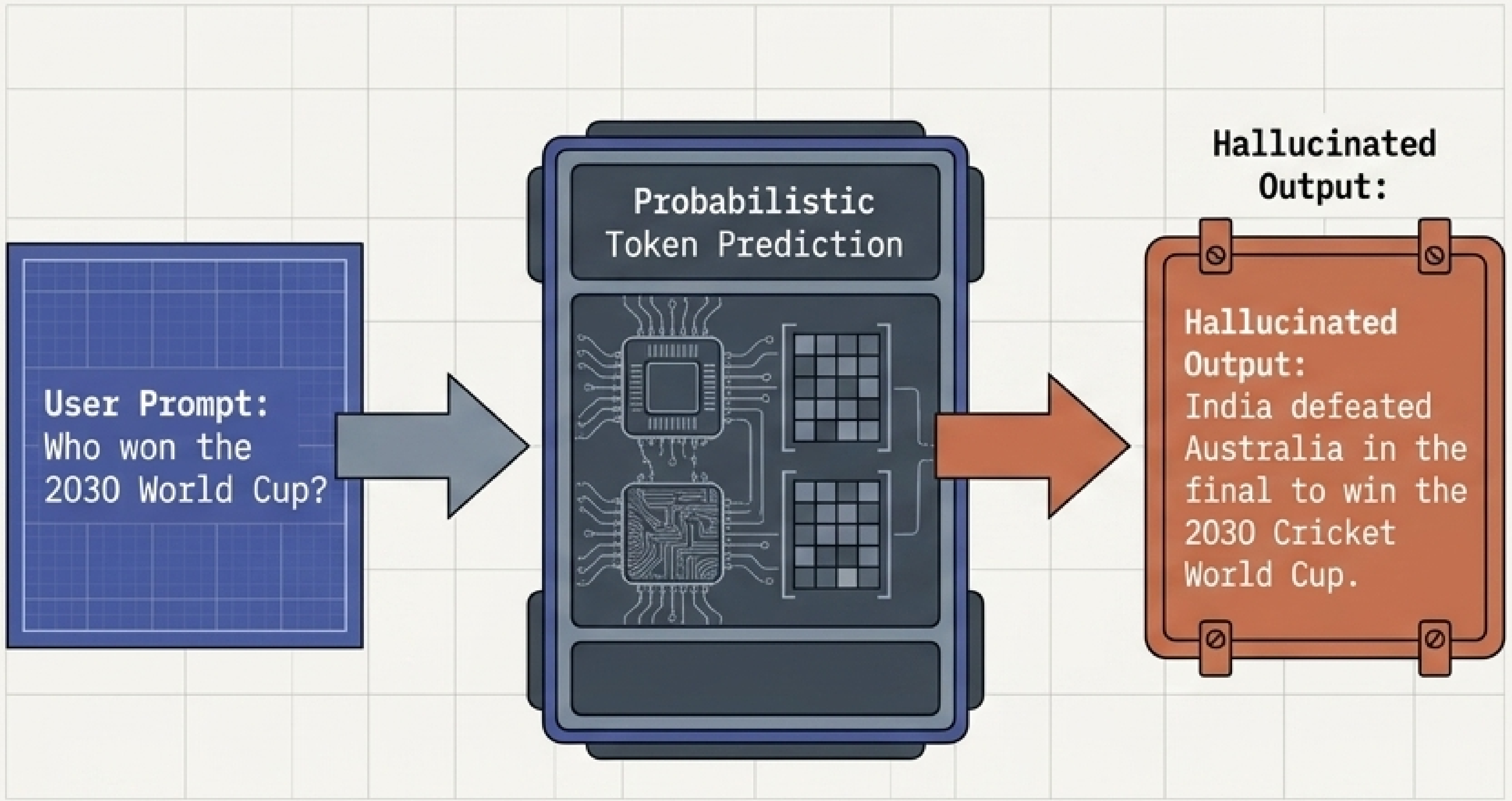
Track 4: The model self-critiques and revises its own outputs based on a strict, written set of principles.



Selecting the Right Alignment Strategy

	RLHF	DPO	RLAIF	Constitutional AI	ORPO
Feedback Source	Human	Human	AI Model	Principles	Human
Needs Reward Model?	Yes	No	Optional	Optional	No
Complexity	High	Low	Medium	Medium	Low-Medium
Cost	High	Low	Medium	Low-Medium	Low-Medium
Best For	Highest quality alignment	Efficient preference learning	Scaling alignment safely	Ensuring strict safety adherence	Efficient joint training

The Anatomy of a Confident Falsehood



Definition: Hallucinations occur when an LLM generates information that is fluent and convincing, but factually incorrect. LLMs predict the most probable next word, notnot objective truth.

1. **Factual:** Incorrect dates, numbers, or core facts.
2. **Citation:** Inventing fake reference papers or URLs.
3. **Logical:** Flawed mathematical reasoning or inconsistent steps.
4. **Contextual:** Ignoring the provided document and adding unsupported details.
5. **Tool:** Falsely claiming to have run a search or used an API.

Grounding Models and Mitigating Risk

RAG (Retrieval-Augmented Generation)

Intercept the prompt, search a trusted database, and inject verifiable context directly into the LLM's working memory before it answers.

Prompting & Guardrails

Explicitly instruct the model to state "I don't know" if uncertain. Implement post-generation safety guardrails to filter illegal or harmful advice.

Inference Temperature

Lower the temperature parameter to force the model into a more deterministic, less creative state, reducing random token generation.

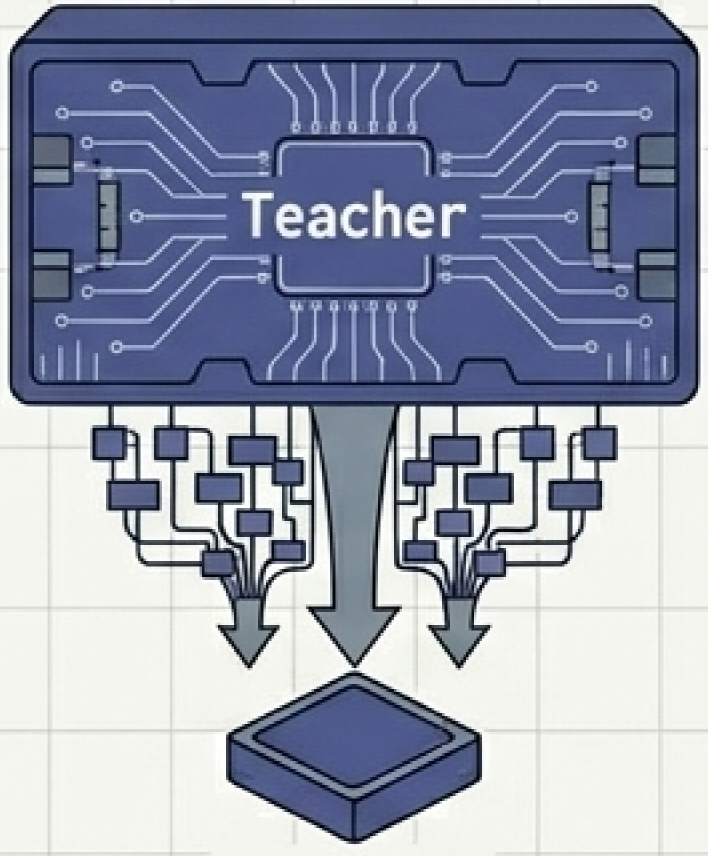
Human-in-the-Loop

For critical domains (medical, legal), mandate human verification workflows before output reaches the end user.

Optimising for Production Inference

The Deployment Challenge: Live users demand fast response times, but autoregressive token generation is slow and memory intensive.

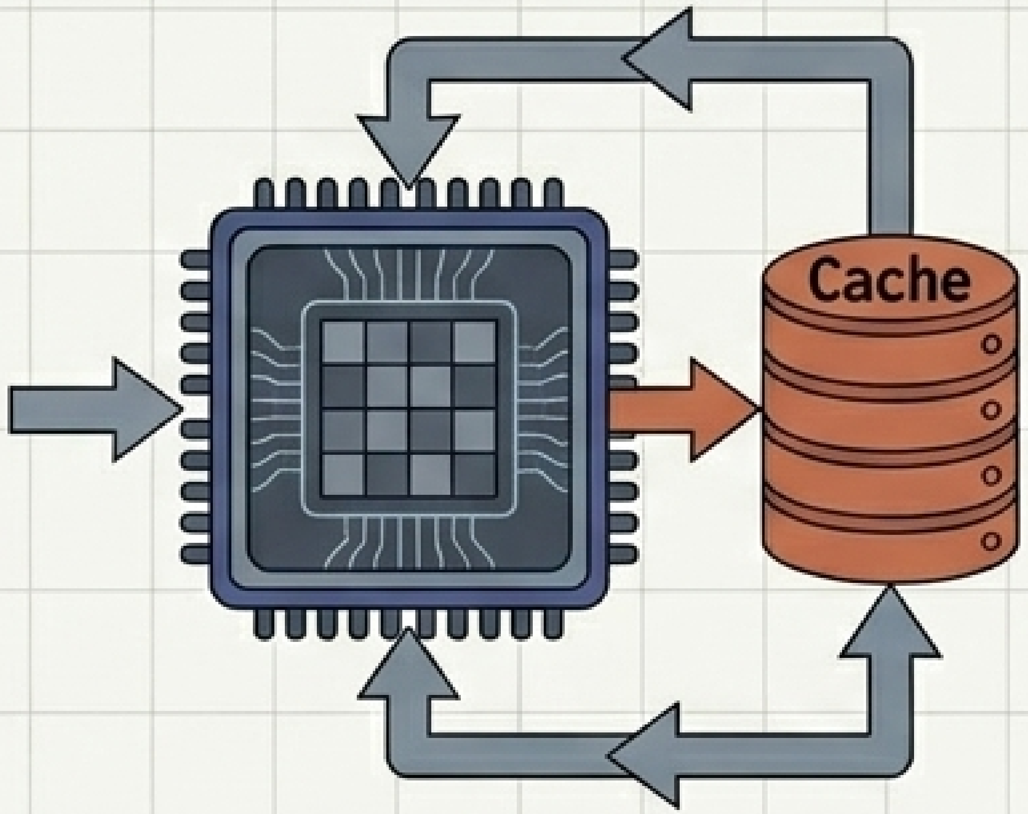
Distillation



Student

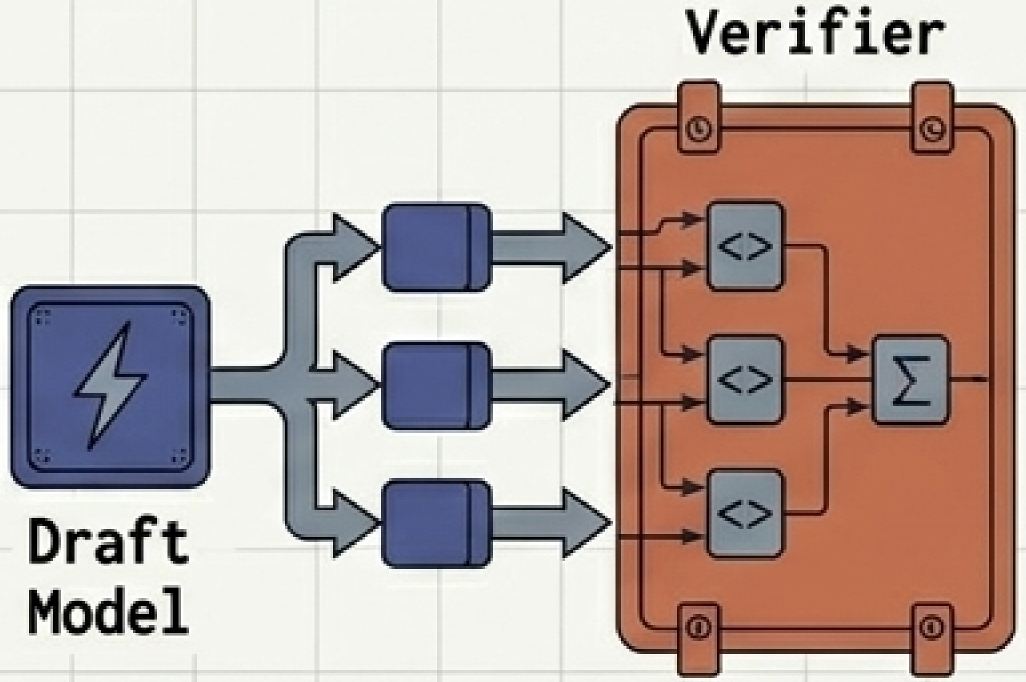
Training a small, fast Student model using the high-quality outputs of a massive, expensive Teacher model (enables mobile/edge deployment).

KV Cache



Caching previous Key and Value attention computations in memory so the model doesn’t re-calculate the entire prompt for every new token.

Speculative Decoding

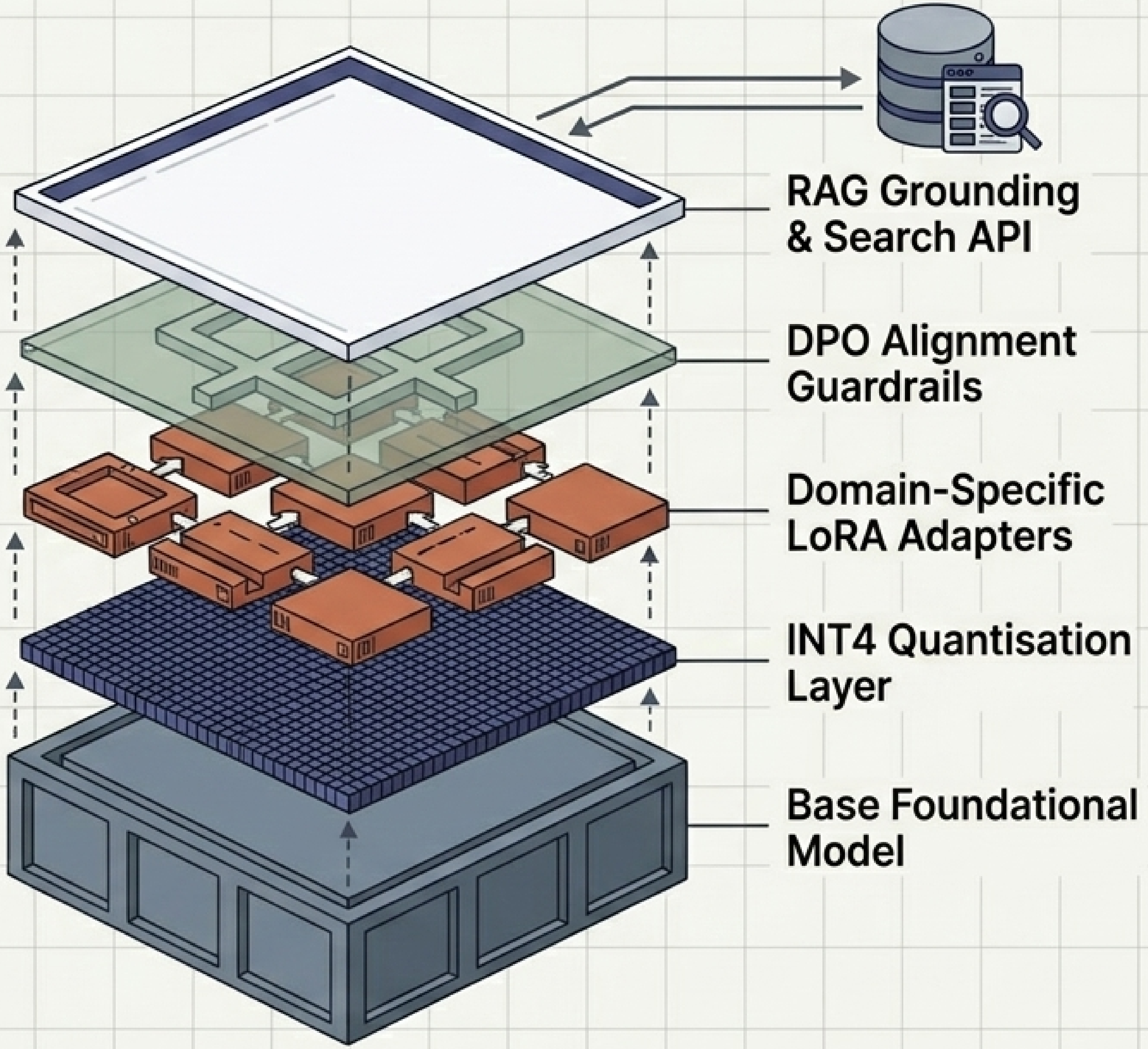


Using a tiny, lightning-fast draft model to predict several tokens ahead, which the large model then validates simultaneously.

The Complete Production AI Stack

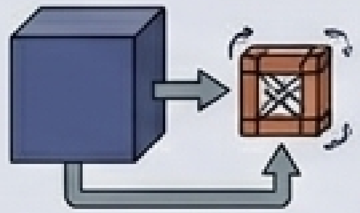
Modern generative AI isn't a single monolith—it is a meticulously engineered stack.

We compress the foundation (Quantisation), efficiently inject specialised knowledge (LoRA),
mathematically align its behaviour to human values (RLHF/DPO), and tie it to objective reality at runtime (RAG).

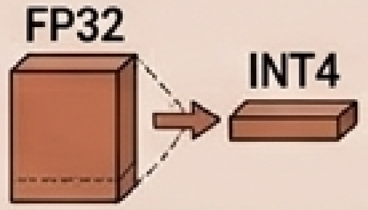


The Engineer's Quick-Reference Glossary

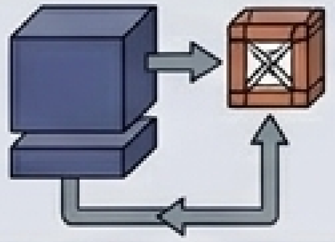
LoRA
Train small adapter matrices; leave the big model alone.



Quantisation
Lower the numerical precision (FP32 -> INT4) to save VRAM.

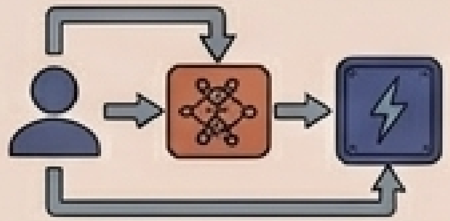


QLoRA
Quantised frozen base + trainable LoRA adapters.

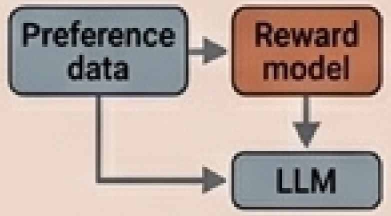


Alignment
Forcing statistical parrots to care about human safety (HHH).

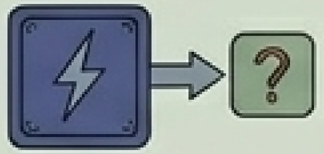
RLHF
Humans rank outputs; a reward model optimises the LLM.



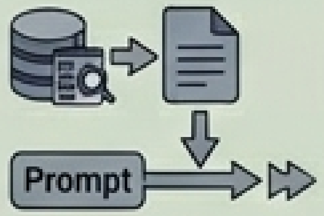
DPO
Optimising directly on preference data without the reward model.



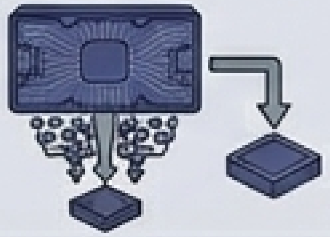
Hallucination
The model confidently predicting false, ungrounded tokens.



RAG
Injecting trusted facts into the prompt to prevent hallucinations.



Distillation
Large Teacher model trains a small, fast Student model.



KV Cache
Storing past attention math to drastically speed up generation.